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Capacity- and Compute-Matched Evaluation of Published BU-Net Components with Multi-Seed Development and Independent External Testing for Multimodal Glioma Segmentation

Article type: Original Research

Summary statement: [PENDING:SUMMARY_STATEMENT_FROM_FINAL_RESULT]

Key Points

1. [PENDING:KEY_POINT_PRIMARY_CAPACITY_CONTROLLED_EFFECT]
2. [PENDING:KEY_POINT_EXTERNAL_FAILURE_OR_DOMAIN_SHIFT]
3. [PENDING:KEY_POINT_RESOURCE_TRADEOFF]

Abstract

Purpose

To test whether the published Residual Extended Skip (RES) component retains a measurable benefit after parameter matching and to characterize uncertainty, lesion-level failure, external domain shift, and measured resource cost.

Materials and Methods

This retrospective public-data study used BraTS 2020 for development and BraTS-Africa for independent external testing. Twelve models were assigned the same five seeds in five patient-level folds. After development-only model, loss, checkpoint, endpoint, and analysis freeze, the confirmatory external cohort was evaluated once. The primary endpoint was patient-level mean whole tumor, tumor core, and enhancing tumor Dice. The primary contrast compared U-

Net+RES with a parameter-matched plain U-Net using paired bootstrap intervals, paired sign-flip tests with Holm correction, and hierarchical seed-patient resampling.

Results

The external confirmatory analysis included [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_PATIENT_COUNT] paired patients. Mean regional Dice was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.FIRST_MEAN] for U-Net+RES and [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.SECOND_MEAN] for the parameter-matched plain U-Net. The paired mean difference was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.MEAN_DIFFERENCE] (95% interval, [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_LOWER_95] to [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_UPPER_95]); the Holm-adjusted P value was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.HOLM_ADJUSTED_P].

Conclusion

The result provides capacity-controlled external evidence about a published BU-Net component and is bounded to the frozen cohorts, models, endpoints, and practical interpretation threshold.

Introduction

Glioma segmentation in multimodal magnetic resonance imaging supports quantitative description of whole tumor (WT), tumor core (TC), and enhancing tumor (ET), but it is also a setting in which evaluation design can dominate an apparent architectural improvement. The BraTS benchmark formalized multimodal subregion segmentation and complementary overlap and boundary metrics [1]. U-Net established an encoder-decoder reference with lateral skip connections [2], followed by residual, contextual, self-configuring, and transformer-based variants.

RES and WC are not new components of the present work. Rehman et al. introduced both in BU-Net: RES transforms information along extended skip pathways, whereas WC uses asymmetric contextual convolutions [3]. Conventional residual blocks have a separate origin in residual learning [4]. Conflating these elements, or comparing a wider component model only with a smaller plain U-Net, cannot isolate why performance differs.

Several additional sources of bias matter. Training stochasticity can change medical-segmentation rankings [5,6]. A self-configuring system such as nnU-Net can outperform manually assembled networks through coordinated choices in preprocessing, training, and inference [7], and controlled reevaluation has shown why modern architectures require strong, fairly tuned baselines [8]. Volumetric and hybrid systems, including TransBTS, nnFormer, and Swin UNETR, also make a purely bounded 2D comparison an incomplete account of the field [9–11].

Metric choice further limits interpretation. Regional Dice does not describe boundary displacement, missed lesions, false-positive lesions, or empty-mask behavior. Metrics Reloaded recommends choosing complementary measures from the task’s problem fingerprint [12].

CLAIM 2024 similarly emphasizes transparent cohort construction, independent evaluation, and complete reporting in medical imaging AI [13].

We therefore framed the work as a controlled evaluation rather than a new-network paper. The primary question was whether RES improves external patient-level mean regional Dice relative to a plain U-Net matched to its parameter capacity. Prespecified secondary analyses evaluated the same-width and compute-matched contrasts, WC and BU-Net configurations, strong baselines, seed-patient uncertainty, lesion and boundary failure, external subgroups, and measured cost. Negative findings were retained by design.

Materials and Methods

Study design and leakage controls

The study was defined before confirmatory external inference. BraTS 2020 training data formed the complete development cohort. BraTS 2019 was used only for identity and duplicate auditing and contributed no additional patient. The legacy internal held-out subset was not reopened for any new model, threshold, loss, post-processing choice, or confirmatory inference.

All splits were patient-level. The [PENDING:DESIGN.DEVELOPMENT_PATIENTS] development patients were assigned deterministically to [PENDING:DESIGN.FOLDS] folds stratified by grade when available, ET presence, and WT burden. Every patient occurred in one validation fold and in no other validation fold. Slice-level random splitting was prohibited.

Development and external cohorts

For each BraTS 2020 patient, the audit recorded identifiers, four modality and segmentation paths, file hashes, NIfTI geometry, affine/orientation, data type, finite-value checks, label set, and regional tumor burden. The expected label set was background, necrotic/non-enhancing tumor, edema, and enhancing tumor. Raw files were read-only and all caches were written outside dataset roots.

The independent cohort was the processed BraTS-Africa TCIA release [14]. Before model inference, each external patient was checked for four modalities, readable labels, label conversion, geometry, and metadata. Identifier, raw hash, normalized content, sampled content, and normalized-volume signatures were compared against all development patients. The primary external cohort contained only glioma; other neoplasms were excluded from confirmatory inference and retained as supportive evidence.

Preprocessing

The input order was T1, contrast-enhanced T1, T2, and FLAIR. Intensities were z-normalized independently by modality over nonzero brain voxels. Spatial augmentation used one transform for all modalities and the segmentation; intensity augmentation was modality-specific. Training sampling could enrich tumor-containing slices or patches, but validation and external evaluation retained the complete volume, including empty slices, so false-positive behavior remained measurable.

Four-class outputs used softmax with the BraTS label mapping. Region metrics were derived as WT = labels {1,2,4}, TC = labels {1,4}, and ET = label {4}. Any nested-region consistency operation was an explicit evaluation stage and was never hidden inside metric computation.

Models and component attribution

The frozen matrix included U-Net-Small; plain U-Nets matched to U-Net+RES by parameters and by declared-input compute; U-Net+RES; U-Net+WC; BU-Net; ResBlock-U-Net; ResBlock-U-Net+WC; official nnU-Net v2 2D; an official hardware-feasible nnU-Net v2 3D full-resolution plan; a five-slice 2.5D U-Net; and MONAI Swin UNETR.

RES and WC followed BU-Net attribution [3]. The ResBlock models used conventional residual mappings sourced to residual learning [4]; they were controls, not novel architectures. The parameter-matched and compute-matched plain U-Nets were chosen by deterministic width/depth search before outcomes. Parameter difference and compute difference were treated as separate constraints rather than collapsed into a subjective score.

Official nnU-Net v2 planning, preprocessing, folds, and trainers were retained where compatible with the frozen data contract [7,8]. The 2.5D comparator stacked five consecutive slices in modality-major order and predicted the center slice, with boundary replication [15]. Swin UNETR used the published hierarchical Swin encoder through the maintained MONAI implementation [11]. No literature score was entered as an experimental observation.

Loss selection and optimization

Loss was selected using development folds only. The candidates were cross-entropy plus soft Dice, binary cross-entropy plus focal Tversky, and cross-entropy plus focal Tversky. Focal Tversky was attributed to Abraham and Khan [16]. The selected objective was **[PENDING:METHOD.SELECTED_LOSS]**. Its executable alpha, beta, gamma, smoothing, channel activation, and background inclusion were stored in the versioned loss catalog and resolved run configuration.

Native 2D models used AdamW with initial learning rate [PENDING:METHOD.INITIAL_LEARNING_RATE_NATIVE_2D], weight decay [PENDING:METHOD.WEIGHT_DECAY], and effective batch size [PENDING:METHOD.EFFECTIVE_BATCH_SIZE_NATIVE_2D]. One warmup-cosine schedule was used; no second scheduler modified the same learning rate. Validation occurred every [PENDING:METHOD.VALIDATION_FREQUENCY_OPTIMIZER_STEPS] optimizer steps.

Convergence runs allowed at most [PENDING:METHOD.MAXIMUM_OPTIMIZER_STEPS] optimizer steps. Early stopping was disabled before [PENDING:METHOD.MINIMUM_OPTIMIZER_STEPS_BEFORE_EARLY_STOPPING] steps and then required a minimum patient-level mean regional Dice improvement of [PENDING:METHOD.EARLY_STOPPING_MINIMUM_DELTA] within [PENDING:METHOD.EARLY_STOPPING_PATIENCE_CHECKS] validation checks. Best and terminal checkpoints were both retained. Compute-matched core runs stopped at a synchronized accelerator budget of [PENDING:METHOD.COMPUTE_MATCHED_HOURS_PER_RUN] hours per run rather than at an equal epoch count.

Every main model used the same [PENDING:DESIGN.SEEDS] seeds in every fold. Failed runs were reported and could not be silently replaced. Model selection, loss selection, and convergence decisions used development artifacts only.

Central evaluation

One evaluator decoded native, MONAI, and nnU-Net predictions and produced patient-level metrics. The primary endpoint was the arithmetic mean of WT, TC, and ET Dice for each patient. Secondary endpoints included regional HD95, surface Dice, sensitivity, precision, specificity, relative volume error, lesion recall, lesion precision, lesion-wise Dice, lesion-wise HD95, and false-positive lesion count. Pixel accuracy was not a primary endpoint.

Lesions were three-dimensional connected components under frozen connectivity and minimum-volume settings. Ground-truth and predicted lesions were paired by a prespecified one-to-one rule. One-empty HD95 remained positive infinity; reports separated its rate from finite medians and interquartile ranges. Evaluation-sensitivity settings changed only connectivity, minimum lesion volume, and reporting caps and did not replace the primary definition.

External evaluation

Gate G required all [PENDING:DESIGN.DEVELOPMENT_RUNS] development runs, required checkpoints, complete validation metrics, frozen loss and nnU-Net plan, environment hashes, and the prespecified statistical inputs. Only its passing freeze authorized Gate H.

Gate H was one external session. It evaluated [PENDING:DESIGN.EXTERNAL_MAIN_CHECKPOINTS] frozen main checkpoints on all external patients without retraining, threshold selection, post-processing selection, or label-informed normalization adaptation. Checkpoint predictions were aggregated within each model by a prespecified nested-region strict majority vote. All checkpoint failures were retained and blocked a passing external gate.

Statistical analysis

The primary contrast was U-Net+RES minus the parameter-matched plain U-Net for external patient-level mean regional Dice. The paired patient bootstrap used [PENDING:METHOD.PAIRED_BOOTSTRAP_RESAMPLES] resamples and confidence level [PENDING:METHOD.CONFIDENCE_LEVEL]. A two-sided paired sign-flip test used [PENDING:METHOD.PAIRED_PERMUTATION_RESAMPLES] Monte Carlo resamples. The primary and four prespecified secondary contrasts formed one Holm family with alpha [PENDING:METHOD.MULTIPLICITY_ALPHA].

The practical interpretation threshold was [PENDING:METHOD.PRACTICAL_THRESHOLD] mean regional Dice. It was not called a clinical minimal important difference and was not an equivalence or noninferiority margin. Effect reporting included the paired mean and median difference, percentile interval, standardized paired effect, adjusted and raw p values, and probability of superiority.

Hierarchical uncertainty resampled training seeds and then patients; folds were averaged in the primary hierarchy and resampled in sensitivity analysis. A mixed-effects model with model and fold fixed effects and patient and seed random intercepts was sensitivity analysis only. External institution, scanner, field strength, grade when available, ET presence, development-based tumor burden, and resolution subgroups were exploratory and reported with explicit denominators and no confirmatory interpretation.

Resource analysis

Parameters, FLOPs, MAC-equivalents, declared input shapes, activation sizes, and a receptive-field proxy were generated from executable model graphs. Training time used synchronized optimizer-step measurements after a frozen warmup. Inference timing separated preprocessing, model forward, post-processing, and end-to-end latency for each patient volume. On Apple MPS, memory was reported as framework-allocated and driver-allocated unified memory, not VRAM.

Accuracy was plotted separately against each cost dimension and in an all-measured-cost non-dominance view. No subjective efficiency score was constructed, and “efficient” was not inferred from parameter count alone.

Qualitative analysis

Case identities were selected after evaluation by six frozen rules: highest patient-average regional Dice, cohort median, lowest finite ET lesion-wise Dice, largest false-positive lesion burden, largest regional HD95, and largest pairwise model disagreement. Ties used larger reference WT volume and then anonymized patient identifier. The language “prespecified cases” was prohibited; only the selection rules were prespecified.

Each panel displayed four modalities, reference labels, reference WT components, predictions from all models, and WT false-positive/false-negative overlays. The axial slice had the largest reference WT area, with the lowest slice index breaking ties. This analysis performed no new inference.

Reproducibility and claim provenance

Each run stored the Git commit, resolved configuration and hash, fold and data hashes, environment and hardware records, random seed, optimizer/scheduler state, best and terminal checkpoint hashes, patient metrics, resource profile, and explicit failure status. Interrupted native MPS runs restored Python, NumPy, and PyTorch CPU RNG state; bitwise claims were restricted to tested contexts.

Gate I rehashed the complete development, external, analysis, table, and figure bundle and reran code checks in a data-free clean clone. Numerical analyses and standard result figures had to regenerate byte-identically. Qualitative panels were hash-verified; their rerender required the ignored derived image cache and was identified as a separate scope boundary. This artifact audit was not described as full retraining.

Gate J generated manuscript values from a machine-readable claim registry. Every substituted value retained its source file hash, row selector, column, format, registry hash, and rendered-document hash. Manual numeric result entry inside artifact-bound Results blocks was prohibited.

Results

Cohort and execution integrity

All [PENDING:DESIGN.DEVELOPMENT_PATIENTS] development patients were assigned to exactly one validation fold. The external confirmatory analysis contained [PENDING:DESIGN.EXTERNAL_CONFIRMATORY_PATIENTS] glioma patients, and the supportive analysis contained [PENDING:DESIGN.EXTERNAL_SUPPORTIVE_PATIENTS] other-neoplasm patients. The complete main external matrix comprised [PENDING:DESIGN.EXTERNAL_MAIN_CHECKPOINTS] checkpoint evaluations.

Primary capacity-controlled comparison

U-Net+RES had mean external patient-level regional Dice [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.FIRST_MEAN], compared with [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.SECOND_MEAN] for the parameter-matched plain U-Net. The paired mean difference was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.MEAN_DIFFERENCE]; the median difference was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.MEDIAN_DIFFERENCE]. The paired percentile interval extended from [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_LOWER_95] to [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PAIRED_BOOTSTRAP_UPPER_95]. The standardized paired effect was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.STANDARDIZED_PAIRED_EFFECT_DZ], the Holm-adjusted p value was

[PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.HOLM_ADJUSTED_P], and the probability of superiority was [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.PROBABILITY_OF_SUPERIORITY]. [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.INTERPRETATION_TEXT]

Resource realization

U-Net+RES contained [PENDING:RESOURCE.UNET_RES.PARAMETER_COUNT] parameters and required [PENDING:RESOURCE.UNET_RES.FLOPS_PER_DECLARED_INPUT] FLOPs per declared input. Mean training time was [PENDING:RESOURCE.UNET_RES.TRAINING_ACCELERATOR_HOURS_MEAN] accelerator hours per fold-seed run, and median external end-to-end latency was [PENDING:RESOURCE.UNET_RES.INFERENCE_END_TO_END_P50_SECONDS] seconds per volume. Peak framework-allocated unified memory was [PENDING:RESOURCE.UNET_RES.TRAINING_PEAK_FRAMEWORK_ALLOCATED_UNIFIED_MEMORY_BYTES_MAX] bytes.

The parameter-matched plain U-Net contained [PENDING:RESOURCE.UNET_PARAMETER_MATCHED_RES.PARAMETER_COUNT] parameters, required [PENDING:RESOURCE.UNET_PARAMETER_MATCHED_RES.FLOPS_PER_DECLARED_INPUT] FLOPs per declared input, and had median end-to-end latency [PENDING:RESOURCE.UNET_PARAMETER_MATCHED_RES.INFERENCE_END_TO_END_P50_SECONDS] seconds per volume. Accuracy-cost non-dominance was reported separately for parameters, FLOPs, training time, allocated unified memory, and end-to-end latency.

Secondary, lesion, subgroup, and qualitative results

All prespecified model contrasts and all frozen models were retained in the generated tables, including negative and failed outcomes. Regional overlap, boundary, empty-mask, lesion detection, lesion-wise, and false-positive lesion endpoints were reported without replacing the primary endpoint. Supportive other-neoplasm results and development-budget sensitivities were separated from the external confirmatory analysis.

Subgroup estimates were exploratory. Cells below the frozen minimum denominator were labeled descriptive, confidence intervals were withheld for those cells, and no multiplicity-adjusted subgroup claim was made. Qualitative panels were chosen by the frozen rules and not by manual case preference.

Discussion

The primary result was: [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.INTERPRETATION_TEXT] This statement is intentionally narrower than a generic claim that residual or context modules improve glioma segmentation. It applies to the exact published RES implementation, its parameter-matched control, the common seeds and folds, the frozen loss and convergence rules, and the independent confirmatory cohort.

The same-width contrast addresses the behavior of the historical base-width comparison but remains capacity-confounded. The parameter-matched contrast is the primary component estimand, whereas the compute-matched contrast tests a different constraint. Agreement among these views would strengthen a component interpretation; disagreement would show that the answer depends on the resource constraint. The study therefore does not collapse them into one ranking.

The strong baselines provide context rather than a tournament assembled from literature scores. nnU-Net tests whether coordinated pipeline configuration outweighs a handcrafted component change [7,8]. The 2.5D and 3D systems test whether through-plane context changes the bounded 2D conclusion [11,15]. Because all predictions pass through one evaluator and one external cohort, differences are not attributed to unmatched datasets or metric code.

Lesion and boundary results are central to interpretation. A favorable mean Dice can coexist with missed ET lesions, infinite one-empty HD95, or false-positive components. We therefore report the complete endpoint family and use qualitative panels to expose success, central tendency, hard failure, false-positive burden, boundary failure, and inter-model disagreement. These panels illustrate measured behavior; they do not estimate prevalence or replace patient-level statistics.

External evaluation on BraTS-Africa introduces population, institution, and acquisition shift that is absent from an internal split [14]. It does not, however, establish universal transportability or clinical utility. The processed cohort and reference labels have their own selection and annotation processes, and the supportive other-neoplasm cases are not part of the confirmatory estimand.

Resource interpretation is similarly bounded. Parameter count, graph compute, training time, allocated unified memory, and latency measure different costs. A model can be favorable on one axis and unfavorable on another. The Pareto views preserve those tradeoffs and avoid converting heterogeneous costs into an arbitrary efficiency score.

Limitations

First, this is a retrospective public-data study and does not evaluate a prospective clinical workflow. Second, the confirmatory cohort is independent of BraTS 2020 by the implemented overlap audit, but it is not representative of every institution, scanner, tumor phenotype, or annotation practice. Third, the primary practical threshold is a prespecified interpretation rule, not a clinical minimal important difference. Fourth, some comparators differ in dimensionality and official training systems; equal patients, folds, seeds, evaluator, and reporting do not make their optimization procedures identical. Fifth, Apple MPS results describe the realized unified-memory device and should not be generalized to CUDA throughput or memory without new measurements. Sixth, artifact-level clean-clone reproduction does not constitute full data-and-compute retraining.

Conclusion

This study evaluates published BU-Net components under patient-level, equal-seed, capacity- and compute-aware development with a single frozen external session. [PENDING:CONTRAST.UNET_RES_VS_UNET_PARAMETER_MATCHED_RES.INTERPRETATION_TEXT] The contribution is controlled evidence and traceability, not a claim that RES or WC was invented here. Clinical utility, universal generalization, and state-of-the-art performance are not established.

Data availability

BraTS 2020 and BraTS-Africa remain subject to their source access terms. Raw medical images and labels are not redistributed by this repository. Versioned manifests, hashes, split assignments, and data-setup instructions are provided to permit authorized reconstruction without exposing source data.

Code and artifact availability

The source code, frozen configurations, tests, and nonrestricted audit artifacts are available at the project repository. A release identifier and archival DOI will be inserted only after the final passing artifact bundle is created. Model checkpoint redistribution is subject to author confirmation and the applicable data-derived artifact terms.

Anonymized Acknowledgments

[AUTHOR CONFIRMATION REQUIRED: enter an anonymized acknowledgment or state none.]

Figure Legends

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References

1. Menze BH, Jakab A, Bauer S, et al. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). *IEEE Transactions on Medical Imaging*. 2015;34:1993–2024. doi:10.1109/TMI.2014.2377694.
2. Ronneberger O, Fischer P, Brox T. U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI*. 2015:234–241. doi:10.1007/978-3-319-24574-4_28.
3. Rehman MU, Cho S, Kim JH, Chong KT. BU-Net: Brain Tumor Segmentation Using Modified U-Net Architecture. *Electronics*. 2020;9:2203. doi:10.3390/electronics9122203.
4. He K, Zhang X, Ren S, Sun J. Deep Residual Learning for Image Recognition. *CVPR*. 2016:770–778. doi:10.1109/CVPR.2016.90.
5. Renard F, Guedria S, De Palma N, Vuillerme N. Variability and reproducibility in deep learning for medical image segmentation. *Scientific Reports*. 2020;10:13724. doi:10.1038/s41598-020-69920-0.
6. Åkesson J, Töger J, Heiberg E. Random effects during training: implications for deep learning-based medical image segmentation. *Computers in Biology and Medicine*. 2024;180:108944. doi:10.1016/j.compbiomed.2024.108944.

7. Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*. 2021;18:203–211. doi:10.1038/s41592-020-01008-z.

8. Isensee F, Wald T, Ulrich C, et al. nnU-Net Revisited: A Call for Rigorous Validation in 3D Medical Image Segmentation. In: *Medical Image Computing and Computer Assisted Intervention—MICCAI 2024*. 2024:488–498. doi:10.1007/978-3-031-72114-4_47.

9. Wang W, Chen C, Ding M, Li J, Yu H, Zha S. TransBTS: Multimodal Brain Tumor Segmentation Using Transformer. In: *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021*. LNCS 12901. 2021:109–119. doi:10.1007/978-3-030-87193-2_11.

10. Zhou H-Y, Guo J, Zhang Y, Han X, Yu L, Wang L, Yu Y. nnFormer: Volumetric Medical Image Segmentation via a 3D Transformer. *IEEE Transactions on Image Processing*. 2023;32:4036–4045. doi:10.1109/TIP.2023.3293771.

11. Hatamizadeh A, Nath V, Tang Y, Yang D, Roth HR, Xu D. Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. In: *BrainLes 2021*. 2022:272–284. doi:10.1007/978-3-031-08999-2_22.

12. Maier-Hein L, et al. Metrics Reloaded: recommendations for image analysis validation. *Nature Methods*. 2024;21:195–212. doi:10.1038/s41592-023-02151-z.

13. Tejani AS, Klontzas ME, Gatti AA, et al. Checklist for Artificial Intelligence in Medical Imaging: 2024 Update. *Radiology: Artificial Intelligence*. 2024;6(4):e240300. doi:10.1148/ryai.240300.

14. Adewole M, Rudie JD, Gbadamosi A, et al. The BraTS-Africa Dataset: Expanding the Brain Tumor Segmentation Data to Capture African Populations. *Radiology: Artificial Intelligence*. 2025;7(4):e240528. doi:10.1148/ryai.240528.

15. Avesta A, Hossain S, Lin M, Aboian M, Krumholz HM, Aneja S. Comparing 3D, 2.5D, and 2D Approaches to Brain Image Auto-Segmentation. *Bioengineering*. 2023;10(2):181. doi:10.3390/bioengineering10020181.
16. Abraham N, Khan NM. A Novel Focal Tversky Loss Function With Improved Attention U-Net for Lesion Segmentation. *IEEE ISBI*. 2019:683–687. doi:10.1109/ISBI.2019.8759329.

Tables

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